

Question:

Four days after undergoing an elective total hip replacement, a 65-year-old woman develops a DVT that embolizes to the lung. Along with tachypnea, tachycardia, and cough, the patient would most likely present with a PaO2 of what?Answer Choices:A. 120 mmHgB. 110 mmHgC. 85 mmHgD. 60 mmHg

Reasoning + Answer:

<think>###FindingReasoningPaths: 1 . **DVT->Thrombusembolism->Pulmonaryembolism->Hypoxemia->DecreasedPaO2 . **2 . **Pulmonaryembolism->Reducedlungperfusion->Decreasedoxygenation->Hypoxemia->DecreasedPaO2 . **3 . **DVT->Secondaryvasoconstriction->Pulmonaryhypertension->Hypoxemia->DecreasedPaO2 . **4 . **Pulmonaryembolism->Acutelunginjuryorinfarction->Reducedlungcomplianceorventilationperfusionmismatch->DecreasedPaO2 . **5 . **Lungembolism->Differentiasignsofoxygenationimpairment->Anaphylaxis (duetothromboembolicreaction) ->Hypoxemia->DecreasedPaO2 . **---###ReasoningProcess: 1 . **DVTandPulmonaryEmbolism:** - Thepatient developeda DVTfourdaysafterundergoingatotalhipreplacement, acommonriskfactorforthromboemboliceventssuchaspulmonaryembolism (PE). - Apulmonaryembolismcausedbya pulmonaryembolus (PE) reducestheefficiencyofoxygenexchangeinthelungs , leadingtohypoxemia . Thisisduetoreducedlungperfusion , decreasedlungcompliance , orthecreationofa ventilation-perfusionmismatch . - HypoxemiainthiscontextmanifestsasadecreaseinPaO2 , typicallyduetosystemichypoxemiaratherthanisolateddiffusepulmonaryhemorrhage . - TheclinicapresentationoftachypneaandtachycardiafurthersupportsthedagnosisofpulmonaryembolismasthePaO2isexpectedtodecreasedespitetheheartrateincrease . 2 . **PulmonaryEmbolismandHypoxemia:** - APEleadstohypoxemiaduetodecreasedlungperfusionandimpairedoxygenexchange . ThisresultsindecreasedPaO2 . - Thedifferentiadiagnosisofhypoxemiadoesnotsuggestisolateddiffusepulmonaryhemorrhageorcirculatoryfailureastheprimarycause . - Thedyspneacouldbeduetodecreasedbloodoxygensupplyorventilation-perfusionmismatchratherthanhypovolemiaorhypoglycemia . 3 . **SecondaryEffectsofPulmonaryEmbolism:** - APEcantriggerasystemicinflammatoryresponse , includingeffectsonbloodcoagulation . Whilethiscouldtheoreticallyleadtoacytokinestorm , theclinicamanifestations(e.g. , dyspnea , tachycardia) aremostdirectlyrelatedtooxygenationimpairmentratherthansystemicinflammation . - HypoxemiaisadirectresultofpulmonaryobstructionratherthanapathologicalgaininPaO2 . 4 . **CirculatoryFailurevs . Hypoxemia:** - CirculatoryfailurewouldcausecyanosisandaparadoxicalincreaseinPaO2duetoreducedsystemicoxygendemand . Thisis inconsistentwiththepatient'spresentationofhypoxemiaandtachypnea . - Hypoxemiaisamoreplausibleexplanationforthepatient'ssymptoms , giventhepulmonaryembolismanditseffectsonlungperfusion . 5 . **PaO2inHypoxemia:** - HypoxemiaischaracterizedbydecreasedPaO2 . Thisisexpectedtooccurinthecontextoflungembolismduetoreducedoxygen delivery , lungperfusion , orventilation-perfusionmismatch . ---###Conclusion: Thepatientis likelyexperiencing** decreasedPaO2 **dueto** hypoxemia **resultingfrom** pulmonaryembolism ** . Thisisconsistentwiththeclinicapresentationoftachypnea , tachycardia , anddyspnea . </think><answer>85mmHg</answer>